

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) aim to reveal how small input changes flip a model’s prediction, yet many methods modify more features than necessary, reducing clarity and actionability. This paper makes four contributions toward CE that are sparse, faithful, and broadly applicable. *First*, we introduce *COLA*, a model- and generator-agnostic post-hoc framework that refines *any* given CE by computing a coupling via optimal transport (OT) between factual and counterfactual sets and using it to select a minimal set of edits while preserving the target effect. *Second*, we propose *p-SHAP*, a joint-distribution-informed Shapley attribution that drives this edit selection and that unifies commonly used Shapley methods as special cases under different couplings. *Third*, we provide theoretical guarantees: OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and under mild conditions the refined counterfactuals are guaranteed not to move farther from the factuals than the originals. *Fourth*, across four datasets, twelve models, and five CE generators, we show empirically that COLA attains the same target effects with only 26–45% of the original feature edits, and that it is near-optimal against a small-scale exact benchmark. Together, these contributions deliver the first framework that systematically minimizes CE edits without committing to a specific CE algorithm or model class.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) is essential for making artificial intelligence systems transparent and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Two families of techniques anchor the setting our contributions build on. The first is feature attributions (FA): methods such as Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) quantify how much each input feature contributes to a machine learning (ML) model’s output, simplifying complex models by highlighting the most influential features. In a healthcare model, for instance, Shapley values can surface key factors like age and medical history to help clinicians understand a decision (Ter-Minassian et al., 2023; Nohara et al., 2022). The second is counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), which show how small changes in input features can lead to different outcomes. Crucially for this paper, no single CE algorithm suits all use cases: despite hundreds of proposals (Guidotti, 2022; Verma et al., 2020), each is tailored to its own scenario, goal, and task. Some seek a single counterfactual per factual instance; others treat instances as a group and return one or multiple counterfactuals per observation. Some target the entire dataset, producing global CE that indicate a direction to move factual instances toward the desired output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024); others view the factual instances as a distribution and seek a counterfactual distribution of comparable shape and cost (You et al., 2025). Some assume differentiable models, while others target tree-based or ensemble models. This heterogeneity is precisely why we design a framework that refines the output of *any* such generator rather than competing with them.

Problem Description and Challenges Our starting point is a gap that neither FA nor CE closes alone. Both are vital for making AI models interpretable and accountable, yet each is incomplete on its own: FA alone rarely yields tangible, actionable steps, while CE alone often prescribes unnecessary feature changes that are impractical to enact. We therefore address a general and comprehensive

problem that couples the two, building on the extensive foundations established in the literature (see Appendix A). Concretely, we seek to answer the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{\$}$	$\textcircled{\text{A}}$	$\textcircled{\text{R}}$	$\textcircled{\$}$	$\textcircled{\text{A}}$	$\textcircled{\text{R}}$	$\textcircled{\$}$	$\textcircled{\text{A}}$	$\textcircled{\text{R}}$	$\textcircled{\text{R}}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] The problem this paper solves: reaching a target outcome with the fewest feature edits. A platform aims to increase user registrations and has collected data on user interactions—money spent ($\textcircled{\$}$), number of clicks ($\textcircled{\text{A}}$), and whether the user has registered ($\textcircled{\text{R}}$). In the original data ($\mathbf{x}$), no users are registered. Action plans $\mathbf{z}'$ and $\mathbf{z}''$ both adjust user characteristics to reach the desired outcome ($\mathbf{y}^*$) of full registration and both achieve a half counterfactual effect, yet $\mathbf{z}''$ does so with fewer modifications than $\mathbf{z}'$. Our contribution COunterfactuals with Limited Actions (COLA) systematically produces the sparser $\mathbf{z}''$ -type plan for any model and CE generator.

Addressing this question raises three challenges, and each directly motivates a design choice we adopt. *Challenge 1 (generality)*: it is unrealistic to expect one CE algorithm to meet all needs, since the problem is inherently task-specific—motivating a generator-agnostic framework. *Challenge 2 (model-agnosticism)*: the approach must not rely on strong model assumptions (e.g., differentiability or special structure), so that it applies across a wide range of models. *Challenge 3 (coupling attribution to the counterfactual)*: standard FA such as feature importance can be misleading, because FA scores need not cohere with the changes required for a counterfactual effect. Put differently, performing FA independently of CE to pick which features to change is ineffective. We show later, in Result II of Section 6, that this decoupling yields counterproductive feature modifications (the “actions” asked of users/stakeholders), since features deemed important in general need not align with the specific pathways that achieve the desired counterfactual outcome. These three challenges are exactly what our contributions, introduced next, are built to overcome.

Main Contributions This paper makes four contributions, each addressing one of the challenges above.

- **(C1) A general post-hoc framework, COLA.** We introduce COLA, which refines CE *across models and across CE generators* by using an optimal transport (OT)–induced coupling between factual and counterfactual sets to guide edit selection. It treats the underlying model and CE generator as black boxes, directly answering Challenges 1 and 2.
- **(C2) A new attribution method, joint-probability-informed Shapley (p -SHAP).** The coupling drives p -SHAP, a joint-distribution-informed Shapley attribution that couples FA to the counterfactual (Challenge 3). p -SHAP *unifies* commonly used Shapley methods as special cases under appropriate couplings and exposes a modular interface for attribution and edit selection.
- **(C3) Theoretical guarantees.** We prove that OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and that under mild conditions the refined counterfactuals are provably *no farther* from the factuals than the originals.
- **(C4) Comprehensive empirical evidence.** Across four datasets, twelve models, and five CE methods, COLA with p -SHAP achieves the same target effects using only 26–45% of the original feature edits, and is near-optimal against an exact benchmark on a small-scale problem. En route, we surface a counter-intuitive finding: even a *known* exact factual–counterfactual alignment can underperform the OT coupling.

Taken together, and to our best knowledge, this is the first method that systematically addresses the problem in Figure 1 *without committing to particular CE algorithms or models*.

2 PROBLEM FORMULATION

We now formalize the problem of Figure 1; this optimization is the common target that all four contributions serve. Denote $f : \mathbb{R}^d \rightarrow \mathbb{R}$ a black-box ML model. Let $\mathbf{x}$ be any observed (factual) data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, $d \geq 1$), and let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The goal is to find a (group of) counterfactual instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) that brings the model output(s) as close as possible to $\mathbf{y}^*$ while spending at most C feature changes—i.e., sparsity is a hard budget, not an afterthought. Let D denote a divergence function measuring dissimilarity between two entities. The problem reads as follows.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The formulation cleanly separates the two ingredients our contributions must reconcile. The objective equation (1a) together with the constraint equation (1b) is the typical CE optimization: $\mathbf{z}$ should make $f(\mathbf{z})$ close to $\mathbf{y}^*$ while staying close to $\mathbf{x}$, so that $\mathbf{z}$ serves as a contrastive reference explaining why $f(\mathbf{x})$ does not reach $\mathbf{y}^*$. We deliberately keep D general—it may be Euclidean distance, OT, maximum mean discrepancy (MMD), or a mean/median difference of the model outcome—so the framework is not tied to any single notion of dissimilarity. The remaining constraints equation (1c)–equation (1e) inject the action budget that makes the problem about *minimality*. Beyond $\mathbf{z}$, we optimize an indicator $\mathbf{c}$ where z_{ik} may change iff $c_{ik} = 1$, with at most C changes allowed by equation (1c). Inspecting equation (1d)–equation (1e): when $c_{ik} = 0$ we have $z_{ik} = x_{ik}$ (no change at (i, k)), and when $c_{ik} = 1$ the sufficiently large constant M gives z_{ik} full freedom to change.

Our route to solving equation (1) is to use FA to identify the most influential features, thereby setting the modification indicator $\mathbf{c}$. The next section reviews commonly used Shapley methods for FA that, together with our proposed p -SHAP (C2), are integrated into the COLA framework (C1) to produce the refined counterfactual $\mathbf{z}$. This design choice is not merely convenient: the problem is computationally hard even when $d = 1$ for linear models (Appendix B), which is why an efficient attribution-guided refinement is needed rather than an exact solver.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews the commonly used Shapley value methods for FA that serve as the baselines our contribution p -SHAP (C2) later generalizes and, in the experiments, outperforms. We first recall the game-theoretic notion of a Shapley value and then survey the standard FA instantiations; (Sundararajan & Najmi, 2020) offers a comprehensive overview. Reading this section as the backdrop for p -SHAP makes the eventual unification (Section 3 onward) easy to track: each baseline below will reappear as a special case of p -SHAP under a particular coupling.

In cooperative game theory, a coalitional game is characterized by a finite set of players (in our context, features), denoted by $\mathcal{F} = \{1, 2, \dots, d\}$, and a characteristic function v . This function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ maps each subset $S \subseteq \mathcal{F}$ to a real number $v(S)$, representing the total payoff that can be achieved by the members of S through cooperation, with the condition $v(\emptyset) = 0$. The Shapley value, as defined based on v later, is a fundamental concept in this framework, providing an equitable method to distribute the overall payoff, $v(\mathcal{F})$, of the grand coalition among the individual players.

Prior to defining the Shapley value, denote $\Delta(k, S)$ the incremental value produced by a player (feature) k to a coalition S , aligning with the concept of marginal contribution. This is defined as the additional payoff realized by the coalition due to the inclusion of the player k , i.e.,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

Intuitively, ϕ_k averages the normalized marginal contributions $\Delta(k, S)$ of feature k over all subsets S that exclude k , with the inverse binomial coefficient $\binom{d-1}{|S|}$ as the normalization factor and d the number of features. Summing over subsets and dividing by d yields the fair, axiomatic distribution of the payoff $v(\mathcal{F})$ among features. These axioms are what our p -SHAP will be shown to inherit later, so the choice of Shapley as the attribution backbone is deliberate rather than incidental.

For an arbitrary data point $\mathbf{x}_i$, $\mathbf{x}_{i,S}$ denotes the part containing only the features in S and $\mathbf{x}_{i,\mathcal{F}\setminus S}$ the remaining part (analogously for other notations). We now review three commonly used Shapley methods; each differs only in how it references a “baseline”, and each will resurface as a special case of p -SHAP.

Baseline Shapley (B-SHAP). This approach uses a fixed baseline point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$, with set function

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F}\setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where a feature’s absence is modeled by its value in the baseline $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). Its defining assumption—an *exact* alignment between $\mathbf{x}$ and $\mathbf{r}$ —is the strongest of the three, and is precisely the assumption our experiments (C4) will show can be suboptimal.

Random Baseline Shapley (RB-SHAP). A variant of B-SHAP, used implicitly in (Lundberg & Lee, 2017) and the widely adopted “SHAP” library, replaces the single baseline with an expectation over a background distribution:

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F}\setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

Here $\mathcal{D}$ is the background distribution, commonly the training set (Lundberg & Lee, 2017; Merrick & Taly, 2020). Notably, $\mathcal{D}$ carries *no* counterfactual information—a gap that motivates using CE-aware couplings and that our results (C4) tie directly to poor sparsity.

Counterfactual Shapley (CF-SHAP). This extends RB-SHAP by taking $\mathcal{D}$ to be the counterfactual distribution defined on each factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$):

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F}\setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

Adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) with demonstrated advantages for contrastive FA, it assumes a *probabilistic* alignment between $\mathbf{x}$ and $\mathbf{r}$. As we make explicit next, B-SHAP, RB-SHAP, and CF-SHAP differ only in the coupling they implicitly assume—the observation that our contribution C2 turns into a single unifying attribution.

4 THE PROPOSED p -SHAP AND ITS THEORETICAL ASPECTS

This section develops contributions C2 (the p -SHAP attribution) and C3 (its theoretical guarantees).

(Contribution C2) p -SHAP Our first technical contribution unifies equation (4) through equation (6) into a single attribution by integrating an algorithm A_{Prob} that returns the joint probability of $\mathbf{x}$ and $\mathbf{r}$.¹ Our p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F}\setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

The key conceptual shift—and the heart of C2—is this: unlike prior methods that rely on random baselines, p -SHAP *solves* the alignment problem by identifying the optimal coupling via OT. This reframes feature attribution not as a mere prediction decomposition but as a transport problem that minimizes the cost of explanation, which is exactly the sparsity objective we care about.

A property central to C1’s generality is that A_{Prob} is *independent of the CE algorithm and depends only on the generated counterfactual $\mathbf{r}$* . By relying solely on these fixed components, p -SHAP keeps

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

FA consistent across the variability of different CE generation processes—a major departure from CF-SHAP and the reason COLA can wrap *any* generator. Contrary to common expectations, we demonstrate that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by [Albini et al. \(2022\)](#), in Result II of Section 6 later—this is the counter-intuitive finding highlighted in C4.

The concrete instantiation we advocate uses the OT problem (equivalently, the 2-Wasserstein divergence) below, and takes the resulting transportation plan $\mathbf{p}_{\text{OT}}$ as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ inside p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions (all admissible transport plans). The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularization with coefficient $\varepsilon \geq 0$; setting $\varepsilon > 0$ accelerates the computation of OT. This coupling is what supplies p -SHAP with a CE-aware alignment at negligible cost.

(Contribution C3) Theoretical Aspects of p -SHAP Our second technical contribution is a pair of guarantees that justify why the OT coupling is the right one. The core claim is simple to state: OT minimizes the total feature-modification cost (moving $\mathbf{x}$ toward $\mathbf{r}$) under the alignment it induces, which corresponds exactly to our objective of achieving the counterfactual outcome at minimal modification cost. We sharpen this into a formal bound under a Lipschitz-continuity assumption on f . Theorem 4.1 below (proof in Appendix C) states that the plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$: the 1-Wasserstein distance $W_1(f(\mathbf{x}), \mathbf{y}^*)$ is bounded by the Lipschitz constant times the square root of the minimized expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where p_{ij} ($j = 1, 2, \dots, m$) quantify how $\mathbf{x}_i$ should be adjusted toward one or multiple $\mathbf{r}_j$. The practical takeaway of C3 is that $\mathbf{p}_{\text{OT}}$ prescribes how to shift $\mathbf{x}$ toward $\mathbf{r}$ at minimal expected cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$: compared to other modification plans $\mathbf{p} \in \Pi$, $\mathbf{p}_{\text{OT}}$ attains the smallest possible cost—hence the *tightest* upper bound on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^)$, where the upper bound is based on the expected feature modification cost.*

The value of this bound is that it acts as a convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem: by concentrating attribution mass on the most efficient paths, the OT coupling guides the greedy selection in Algorithm 1 to discard non-essential features—turning C3 into the engine behind C1’s sparsity. A second guarantee shows that p -SHAP is conceptually correct in attributing behavior to feature modifications, stated in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Finally, completing C3, p -SHAP preserves the desirable axioms of B-SHAP and RB-SHAP, so unifying them (C2) costs nothing in attribution soundness; we omit the proof and refer to ([Sundararajan & Najmi, 2020](#); [Lundberg & Lee, 2017](#)) for the Shapley axioms.

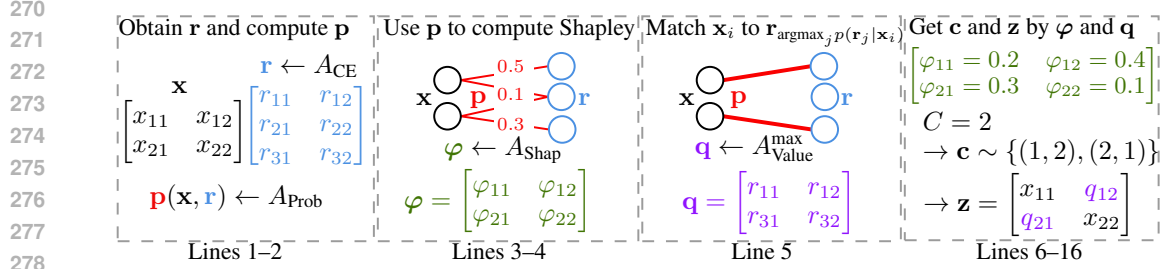


Figure 2: [An illustration of COLA] End-to-end view of how COLA (contribution C1) turns any counterfactual into the sparse action plan $\mathbf{c}$ and refined counterfactual $\mathbf{z}$ for equation (1). For clarity we use $A_{\text{Value}}^{\text{max}}$ in line 5. In lines 6–16 we set $C = 2$, so sampling according to the attribution matrix ϕ (produced by p -SHAP) selects exactly two positions to modify.

5 THE ALGORITHMIC FRAMEWORK COLA

This section presents contribution C1: the COLA framework that operationalizes p -SHAP into a working refinement procedure.

Sketch COLA (Algorithm 1) solves equation (1) by composing four interchangeable categories of algorithms—the modularity that makes it generator- and model-agnostic (Challenges 1–2). We detail Algorithm 1 below, with an illustration in Figure 2; note that only Line 3 carries our new attribution (C2), while the surrounding steps are deliberately generic so that any CE generator can be plugged in.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \{0, 1\}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\phi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\phi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{m \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix ϕ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). Any CE algorithm A_{CE} takes the model f , factual $\mathbf{x}$, target $\mathbf{y}^*$, and tolerance ϵ , and returns a counterfactual $\mathbf{r}$ that stays close to $\mathbf{x}$, with the returned $\mathbf{r}$ satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$. This is the sole point of contact with the underlying generator, and treating it as a black box is what delivers C1’s generator-agnosticism.

Line 2 (Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$). The algorithm A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ and outputs a matrix encoding the joint distribution over all n factual and m counterfactual points in $\mathbf{x}$ and $\mathbf{r}$. This coupling $\mathbf{p}$ —the crux of C2—represents the alignment (or matching) between the factual rows and counterfactual rows, and is reused in Line 5 to compose $\mathbf{z}$. As discussed in Section 3, when the row alignment between $\mathbf{x}$ and $\mathbf{r}$ is unknown we set A_{Prob} to OT, yielding the coupling of smallest OT distance between $\mathbf{x}$ and $\mathbf{r}$; when a trustworthy alignment is available, one may instead supply it directly.

Lines 3–4 (*p*-SHAP FA—the C2 core). This is where our new attribution enters. We apply equation (7) to compute the Shapley values of $\mathbf{x}$, using the coupling $\mathbf{p}$ to align each row of $\mathbf{x}$ with its corresponding counterfactual rows of $\mathbf{r}$, so that the selected rows in $\mathbf{r}$ serve as the most representative contrastive reference for that row of $\mathbf{x}$; numerically, this alignment is what drives the quality of the contrastive FA. The Shapley matrix of $\mathbf{x}$ is then taken element-wise in absolute value and normalized to sum to one, giving the attribution matrix φ .

Line 5 (*Computing feature values*). The algorithm A_{Value} takes $\mathbf{r}$ and the coupling $\mathbf{p}$ and, for each row i of $\mathbf{x}$, selects one or several rows of $\mathbf{r}$ as references for the changes in $\mathbf{x}$. It returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ whose entry q_{ik} is the counterfactual candidate for x_{ik} . We give two instances— $A_{\text{Value}}^{\text{max}}$ for selecting a single row and $A_{\text{Value}}^{\text{avg}}$ for selecting multiple rows.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\text{max}}$ selects the single row of $\mathbf{r}$ with the highest coupling probability.

$$\mathbf{q} = A_{\text{Value}}^{\text{max}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ instead sets q_{ik} to a convex combination (coupling-weighted average) of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16 (*Sparse edit selection*). Since φ is non-negative and sums to one, we treat it as a selection policy over positions of $\mathbf{x}$: line 7 samples the C positions to edit (setting $c_{ik} = 1$). Each such x_{ik} is then replaced by q_{ik} , and the modified matrix $\mathbf{z}$ together with $\mathbf{c}$ constitutes the solution to equation (1). This is the step where C2’s attribution is converted into the sparse action plan that C1 promises.

By Theorem 4.1, COLA minimizes the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by editing $\mathbf{z}$ according to the attribution, which pinpoints the features most worth adjusting to achieve the counterfactual effect. Completing the C3 guarantees, the theorem below (proof in Appendix E) shows the refined $\mathbf{z}$, produced by the COLA framework, satisfies the proximity constraint equation (1b): in the typical case $n = m$ with the Frobenius norm, refinement never pushes $\mathbf{z}$ farther from $\mathbf{x}$ than the original counterfactual. Table 3 provides empirical evidence that this conclusion holds more broadly.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Complexity of COLA The refinement that C1 adds on top of the CE generator is cheap. Let $\mathcal{O}(M_{\text{CE}})$ be the complexity of A_{CE} , and for A_{Shap} let M_{Shap} be the number of sampled subsets under a weighted-linear-regression estimator. The complexity of COLA in n, m, d , and the entropic-OT regularizer ε is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used. See Appendix F for more details.

6 NUMERICAL RESULTS

This section delivers contribution C4: the empirical evidence that COLA solves equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ and $\mathbf{r}$ obtained from a CE method A_{CE} . We report under four divergence functions:

OT (distance between full distributions), MMD (divergence between distribution means in a high-dimensional feature space), and absolute mean difference (MeanD)/absolute median difference (MedianD) (differences in mean/median). The evaluation is organized around three claims, each mapping to a contribution: I) *COLA achieves modification minimality* (validates C1). II) *p-SHAP outperforms other Shapley methods for modification minimality* (validates C2, and surfaces the counter-intuitive finding in C4). III) *COLA is near-optimal against an exact benchmark* (validates C1/C4).

Experiment Setup To substantiate the breadth claimed in C1 and C4, we span 4 binary-classification datasets, 5 CE algorithms designed for diverse goals, and 12 classifiers (Table 1). A combination of dataset, A_{CE} algorithm, and model defines one “experiment scenario”, giving a wide grid over which the method must hold up.

Table 1: [Experiment Scenarios Setup] The breadth behind contribution C4: four datasets, five CE algorithms, and 12 models. A “scenario” is a combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The 6 compared methods (colors match Figure 3), where CF- p_{OT} is our proposed p -SHAP (C2) and the rest are baselines. The set is designed as an ablation that isolates the effect of the OT coupling; each method is run across the scenarios of Table 1.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

The chosen A_{CE} in Table 1 deliberately cover the three CE paradigms that C1 must handle: DiCE and KNN are instance-based (a counterfactual per factual instance); AReS and GlobeCE are group-based (a shared set for the whole factual group); and Discount is distributional (it treats the factual instances as an empirical distribution and seeks a nearby counterfactual distribution).

Table 2 defines the 6 compared methods, with CF- p_{OT} being our proposed p -SHAP (C2) and the rest baselines. Each method is a specific choice of the three algorithms A_{Prob} , A_{Value} , and A_{Shap} , and each is benchmarked across many scenarios of Table 1. For example, RB- p_{Uni} uses a uniform A_{Prob} to compute $\mathbf{p}$, feeds $\mathbf{p}$ into A_{Value} via equation (9)/equation (10) to get $\mathbf{q}$, and uses RB-SHAP as A_{Shap} .

The comparison is a controlled ablation that pins the benefit of p -SHAP on the OT coupling itself rather than incidental factors—this is what makes the C2 claim credible. (i) RB- p_{Uni} vs. CF- p_{Uni} isolates whether using CE information helps at all. (ii) CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all use CE in FA but with different couplings, isolating the effect of the coupling; we show the OT coupling wins by a wide margin. (iii) CF- p_{OT} vs. RB- p_{OT} (which differ only in A_{Shap}) confirms that OT helps by informing attribution with the factual-counterfactual alignment, not for other reasons. (iv) CF- p_{Ect} is the special case in which each counterfactual has a *known* source, an exact alignment that makes CF-SHAP coincide with B-SHAP—the reference against which the counter-intuitive finding of C4 is measured.

Table 3: [COLA for Modification Minimality (evidence for C1 and C2)] Number of modified features in $\mathbf{z}$ per method when $f(\mathbf{z})$ reaches 80% (and 100%) of the counterfactual effect of $f(\mathbf{r})$; fewer is better. Each entry averages 4 randomly selected scenarios of Table 1 with 100 runs each. “–” marks that the target effect is unreachable. Our p -SHAP (CF- p_{OT} , in bold) is the only method that consistently reaches the targets and does so with the fewest edits.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	–	–	–	–
	RB- p_{OT}	5.29(± 0.09)	75.9%	–	–
	CF- p_{Uni}	–	–	–	–
	CF- p_{Rnd}	–	–	–	–
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	–	–
	RB- p_{OT}	8.55(± 0.08)	50.2%	–	–
	CF- p_{Uni}	7.01(± 0.07)	41.1%	–	–
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	–	–
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	–	–
	RB- p_{OT}	–	–	–	–
	CF- p_{Uni}	2.80(± 0.04)	34.4%	–	–
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	–	–
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	–	–	–	–
	RB- p_{OT}	–	–	–	–
	CF- p_{Uni}	–	–	–	–
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	–	–
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result I (validates C1): COLA cuts actions sharply at minor cost to the counterfactual effect

In Table 3, we ask $\mathbf{z}$ to reach 80% or 100% of the counterfactual effect of $f(\mathbf{r})$ ². The headline number for C1: COLA reaches these targets with far fewer actions in $\mathbf{z}$ (i.e. feature modifications) than the original CE $\mathbf{r}$ —only 13%–25% of the feature changes (measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$) suffice for the 80% target. Moreover, COLA with p -SHAP (CF- p_{OT}) is the *only* method that reaches the full 100% effect, and does so with just 26%–45% of the feature changes in $\mathbf{r}$. This best-in-class action minimality of p -SHAP is exactly what Result II dissects next.

Result II (validates C2): p -SHAP beats the other Shapley methods at achieving the counterfactual effect

Figure 3 plots $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) against the action budget C for the Shapley methods of equation (4)–equation (7). The ablation reads off cleanly, one controlled comparison at a time. *First*, RB- p_{Uni} and RB- p_{OT} trail the rest by a wide margin—using CE information is essential for attribution toward CE sparsity. *Second*, CF- p_{OT} beats RB- p_{OT} , and since the two differ only in A_{Shap} (Table 2), the gain is attributable to the OT alignment feeding p -SHAP, not to other factors. *Third*, p -SHAP markedly outperforms CF- p_{Uni} and CF- p_{Rnd} , showing that merely knowing the counterfactual (and its marginal) is not enough: the *coupling* matters, and—foreshadowing C4—the right coupling need not be the one defined by the exact CE generation mechanism (Figure 4).

These gains for COLA with p -SHAP hold across most scenarios of Table 1 (Figure 3, Table 3), with a large body of further evidence in Appendix H. In short, p -SHAP is the strongest Shapley method for CE sparsity—the core claim of C2.

Result III (validates C1/C4): COLA is competitive with the MILP optimum on tractable benchmarks

This result serves two contributions at once. It quantifies how close COLA comes to the exact optimum (C1), and it delivers the counter-intuitive finding of C4: p -SHAP can eliminate the

²By definition $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, i.e. a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. An 80% counterfactual effect is the proportion of divergence removed by the refined CE $\mathbf{z}$, namely $Counterfactual\ Effect = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

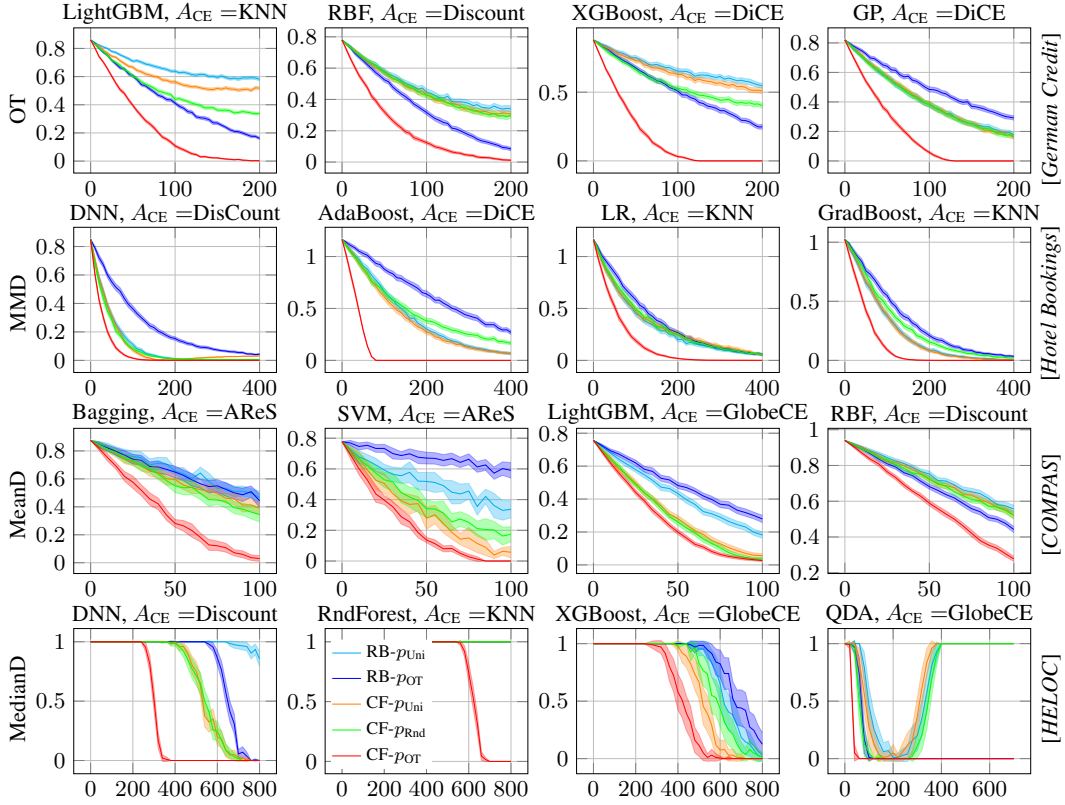


Figure 3: Evidence for C2: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. the action budget C (lower and further-left is better). Across every panel, our p -SHAP (CF- p_{OT} , red) drives the divergence down fastest. Experiments use 100 runs; shaded bands are 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ for *German Credit* and *Hotel Bookings*. The legend in “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

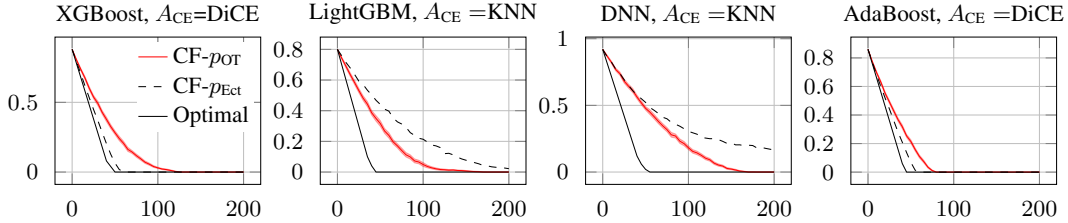


Figure 4: [*German Credit*] Evidence for C4’s near-optimality and its counter-intuitive finding: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. the action budget C with $D = \text{MeanD}$. CF- p_{OT} (our p -SHAP) tracks the exact mixed integer linear programming (MILP) optimum closely and at times matches or beats CF- p_{Ect} , which is granted a known exact alignment.

influence of the CE generation process by replacing the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT coupling, as shown in Figure 4. We benchmark CF- p_{Ect} within COLA on equation (1) with MeanD as D ; because CF- p_{Ect} relies on a known alignment, it tests how well COLA exploits such an alignment. The exact optimum here is obtained by an MILP (derivation in Appendix G); since the MILP is computationally heavy, we solve it only for *German Credit*.

CF- p_{Ect} is near-optimal under DiCE. Because DiCE and KNN produce mutually independent factual-counterfactual pairs, this shows COLA is effective even for instance-based CE, despite equation (1)

³This knowledge—an exact alignment between factual and counterfactual—is available only via DiCE and KNN among the CE methods considered.

being posed for the more general group/distributional setting. The striking observation—the heart of C4—is that $CF-p_{OT}$ *sometimes beats* $CF-p_{Ect}$: the OT coupling identifies the features worth modifying on more theoretically grounded terms, whereas the exact-alignment knowledge used by $CF-p_{Ect}$ has no principled justification for FA. A gap to the optimum remains for $CF-p_{Ect}$ and $CF-p_{OT}$ under KNN, so finding the best coupling is left as an open question—a natural next step beyond the contributions claimed here.

7 CONCLUSIONS

We revisit the four contributions in light of the evidence. **(C1)** We introduced COLA, a model- and generator-agnostic post-hoc framework that refines any CE into a sparse action plan preserving the counterfactual effect. **(C2)** At its core is p -SHAP, a joint-distribution-informed Shapley attribution that unifies common Shapley methods and couples attribution to the counterfactual via an OT alignment. **(C3)** We proved that this coupling minimizes an upper bound on the W_1 divergence to the target and that the refined counterfactual never drifts farther from the factual than the original. **(C4)** Across four datasets, twelve models, and five CE generators, COLA attained the same target effects with only 26–45% of the original edits and stayed near an exact MILP optimum, while revealing that a known exact alignment can be bettered by the OT coupling. Together these establish, to our knowledge, the first framework that systematically minimizes CE edits without committing to a specific CE algorithm or model. Promising next steps include extending the approach to more complex, higher-dimensional models and to settings where a causal graph is available, and searching for couplings that close the remaining gap to optimality.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (COLA with p -SHAP) for refining CE. Our work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses/terms and the common research practice of using these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immutability constraints on features.

CE can surface biases in underlying models but do not by themselves guarantee fairness. Our method can reduce the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age) nor does it confer causal validity. We discourage use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should include domain-expert–defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional (not causal) and must not be interpreted as causal effect.

Risks include (i) misinterpretation of attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these by (a) stating non-causal scope and limitations in the paper, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication; we report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

⁴<https://iclr.cc/public/CodeOfEthics>

REPRODUCIBILITY STATEMENT

We take reproducibility seriously and give pointers to every component behind our contributions. The COLA framework (C1) and the p-SHAP attribution (C2) are specified with pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), and the theoretical guarantees (C3)—all assumptions and complete, line-by-line proofs—are deferred to the appendix.

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also describes the datasets used and the evaluation protocol (including data splits and preprocessing). Baselines (models and CE generators) and coupling choices are enumerated in the experimental sections and ablations, with their configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; we also report hardware and runtime in Appendix I to contextualize compute requirements, and provide ablation/sensitivity studies and the small-scale MILP benchmark setup in the appendix to support independent verification.

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APPENDICES

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A COMPARISON TO EXISTING APPROACHES

We position our contributions against the closest prior work, emphasizing where each of C1–C4 goes beyond it. [Albini et al. \(2022\)](#) proposed CF-SHAP, which uses counterfactual data points as the Shapley background distribution but assumes that distribution is defined conditionally on each single instance—i.e., a *known* probabilistic alignment between every factual and every counterfactual. The same assumption recurs in ([Merrick & Taly, 2020](#); [Kommiya Mothilal et al., 2021](#)) and fails whenever global explanations are sought; the setup in ([Albini et al., 2022](#)) for the counterfactual data distribution is in fact a special case of ours (recovered by C2). [Kwon & Zou \(2022\)](#) proposed WeightedSHAP, weighting features rather than treating them equally; our method instead weights data points and extends naturally to weighting both rows and columns. [Kommiya Mothilal et al. \(2021\)](#) builds a framework combining FA and CE, but its CE-based FA inherits the alignment assumption of ([Albini et al., 2022](#)) and thus does not generalize to group ([Ley et al., 2023; 2022](#)) or distributional CE ([You et al., 2025](#))—precisely the generality C1 targets. [Sharma et al.](#) models counterfactuals via a system’s causal structure and time-series predictors to attribute a single observed metric change to inputs. Most importantly, none of this literature addresses the minimal-action CE problem that is the focus of our paper.

The problems in ([Kanamori et al., 2022](#); [Karimi et al., 2021](#)) come closest to ours, and contrasting them clarifies the value of C1 and C3. [Karimi et al. \(2021\)](#) minimize the cost of actions but assume a known structural causal model (SCM), which is rarely available in practice. [Kanamori et al. \(2022\)](#) note that it remains open whether existing CE methods can solve the problem, and propose an MILP-solver approach for linear classifiers, tree ensembles, and deep ReLU networks, building on ([Ustun et al., 2019](#); [Kanamori et al., 2020](#); [Parmentier & Vidal, 2021](#)). Such MILP solvers, however, are costly and hard to scale—which is why our framework instead uses cheap attribution-guided refinement backed by the guarantees of C3, and reserves the MILP only as an exact benchmark (Result III).

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

This hardness result is what justifies the design behind C1: Theorem [B.1](#) shows that, unless $\mathcal{P} = \mathcal{NP}$, no scalable exact algorithm exists in general, with the difficulty arising not only from the non-linearity of f but also from the combinatorial edit-selection structure. An efficient attribution-guided refinement is therefore not a shortcut but a necessity.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the goal is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error while having at most K non-zero

elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We will map this SR problem to our problem equation (1) with the following settings. Let the number of features be $d = 1$ and the number of instances be n (the same as the dimension of the SR problem). Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ (the zero vector), and the target model output be $\mathbf{y}^* \in \mathbb{R}^m$ (as given in the SR problem). We define the model f as a linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, we define $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$, i.e. the Euclidean distance squared. For the instances, we define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We set the maximum allowed feature changes C to be equal to k (the sparsity level from the SR problem). The large constant M can be any sufficiently large positive number, for example, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

This means that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, the variables c_i indicate whether the variable z_i is allowed to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce that $z_i = 0$ when $c_i = 0$, mirroring the sparsity constraint in the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features can change, matching the sparsity level. The objective function is identical to that of SR.

Since our problem formulation directly mirrors the SR problem with a cardinality constraint, which is known to be $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving the SR problem. Therefore, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergence functions D are standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below—the formal backbone of contribution C3—shows that the attributions align with the expected costs of feature modifications, so the resulting action plans are cost-efficient in achieving counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

In words—and reinforcing the interventional half of C3—the value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ under the intervention that sets the features in S to $\mathbf{x}_{i,S}$, i.e. $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint distribution $p(\mathbf{x}, \mathbf{r})$, so $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ combines the base expected value of $f(\mathbf{r})$ with the additional causal impact of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

To back the efficiency claim underlying C1, we analyze the computational complexity of COLA step by step.

First, we analyze A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} just constructs the matrix $\mathbf{p}$ with the prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise, we consider solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\text{max}}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{max}}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This appendix derives the exact benchmark used in Result III to substantiate the near-optimality claim of C4. It provides a global-optimality reference for solving equation (1) with a known factual–counterfactual alignment and D being MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because $\text{CF-}p_{\text{Ect}}$ is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under the two constraints equation (14b) and equation (14c) is equivalent to minimizing the objective function of equation (13). The constraints in equation (14d) guarantees that each data point i is subject to one and only one feature modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$). The constraint equation (14e) corresponds to equation (13b). Solving equation (14) yields the theoretical optimality of COLA using a known alignment between factual and counterfactual, demonstrated in Figure 4 in Section 6.

H EXTENDED NUMERICAL RESULTS

This appendix expands the evidence for C2 and C4 with the full per-scenario plots. We first study how the choice of Shapley method affects FA (lines 3–4 of COLA), via two controlled comparisons that each hold everything but A_{Shap} fixed. *First*, $\text{RB-}p_{\text{Uni}}$ vs. $\text{CF-}p_{\text{Uni}}$: across Figures 5–8, $\text{CF-}p_{\text{Uni}}$ generally outperforms $\text{RB-}p_{\text{Uni}}$. *Second*, $\text{RB-}p_{\text{OT}}$ vs. $\text{CF-}p_{\text{OT}}$: the latter consistently wins. Both comparisons point to the same conclusion—*RB-SHAP is unsuitable for FA in CE*, reinforcing C2.

The other equally important step is line 5, which uses the coupling $p(\mathbf{x}, \mathbf{r})$ to build $\mathbf{q}$ —the direction in which to move each factual toward the target outcome. Here CF- p_{OT} consistently outperforms every other method across all experiments in Figures 5–8. All three of CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} supply a coupling $\mathbf{p}$ when the exact alignment is unknown, yet their performance differs sharply—again showing (C2) that simply knowing the CE and its marginal is insufficient; the coupling itself is decisive.

OT is especially valuable when the alignment between factual and counterfactual is missing or inaccurate. Even when a CE algorithm explicitly matches each factual to a counterfactual, it is hard to justify that this known alignment is performance-optimal—the counter-intuitive finding of C4, supported by Figure 4 in Section 6.

We stress that p_{OT} need not be the *true* data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$; its role is to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ jointly for both FA and CE. The QDA column of Figure 5 further illustrates the stability of OT-based methods while others diverge from the target. Finally, COLA is *not limited to OT* as A_{Prob} : as Figure 4 shows, even the best known $\mathbf{p}$ leaves a non-negligible gap to global optimality, so designing a stronger A_{Prob} is an open direction that our modular framework (C1) is built to accommodate.

I EXPERIMENTS REPRODUCIBILITY

To make the empirical contribution (C4) reproducible, we document the full experimental environment. Experiments run on a high-performance computing (HPC) cluster with four nodes (one per dataset) in parallel, each node equipped with two Intel Xeon 2660v3 processors (10 cores, 2.60GHz) and 128 GB memory, taking roughly 5–10 hours per node depending on dataset size. The experiments also reproduce on a laptop, though generally at greater computational cost than on the cluster.

Across the four datasets, numerical features are standardized and categorical features use either label or one-hot encoding; we observed no notable difference between the two encodings for COLA’s performance. The train–test split is 7 : 3.

The optimality baseline in Figure 4 (the exact reference for C4) is solved with Gurobi 11.0.2 ([gur](#)), which requires a license; an open-source alternative is Google OR-Tools ([goo](#)). As noted, the MILP of Appendix G is computationally expensive and practical only for small-scale datasets such as German Credit; computing it for other datasets requires reducing the number of features used.

Model hyperparameters are as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interfaces with default hyperparameters. This spread of twelve model families is what lets C4 claim model-agnostic breadth.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

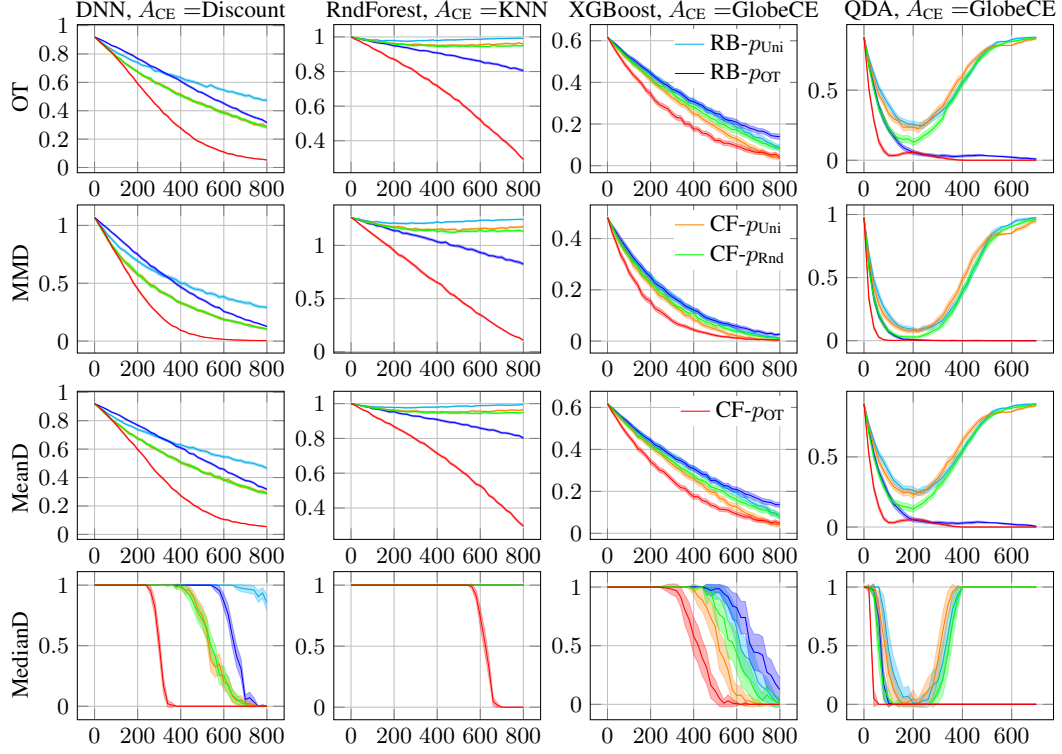


Figure 5: [HELOC] Extended per-scenario evidence for C2: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. the action budget C , where CF- p_{OT} (our p -SHAP) leads throughout. Experiments use 100 runs; shaded bands are 99.9% confidence intervals; legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

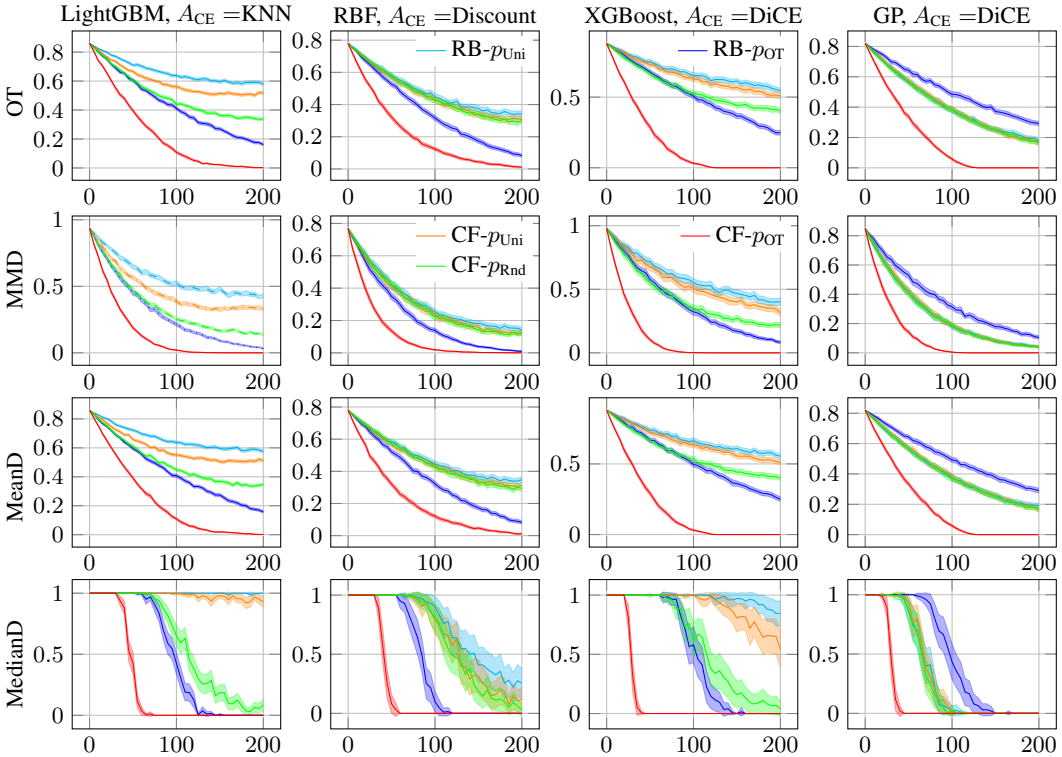


Figure 6: [German Credit] Extended per-scenario evidence for C2: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. the action budget C , with CF- p_{OT} (our p -SHAP) consistently ahead. Experiments use 100 runs; shaded bands are 99.9% confidence intervals; legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

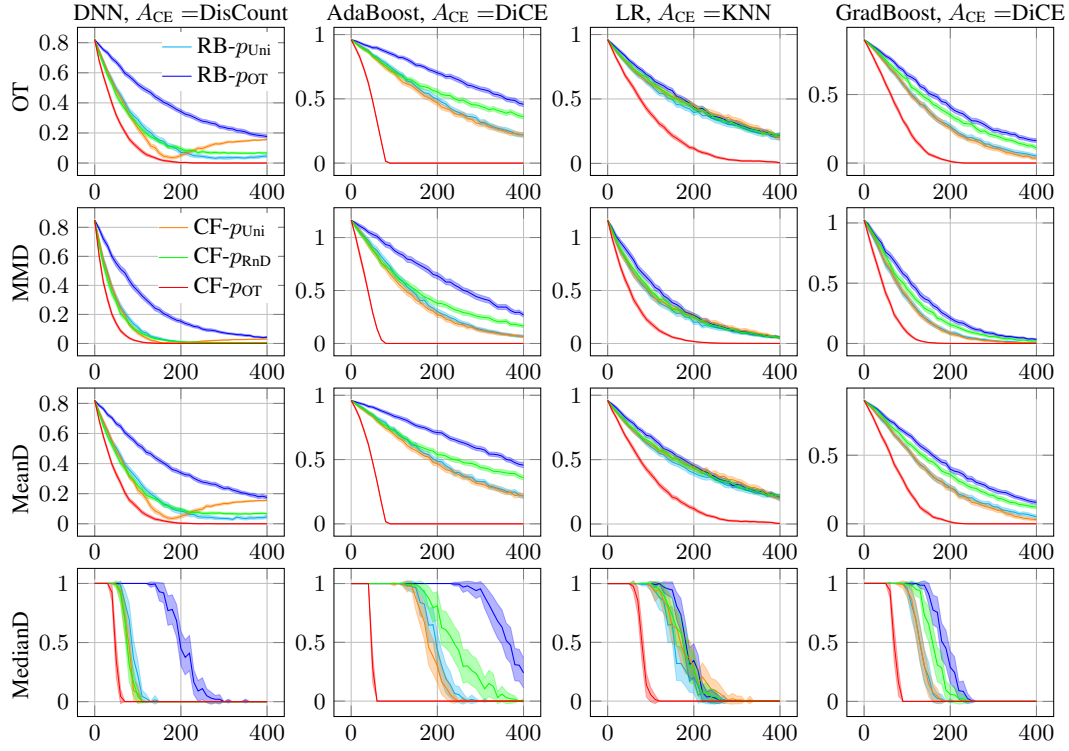


Figure 7: [Hotel Bookings] Extended per-scenario evidence for C2: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. the action budget C , with $\text{CF-}p_{\text{OT}}$ (our p -SHAP) leading throughout. Experiments use 100 runs; shaded bands are 99.9% confidence intervals; legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

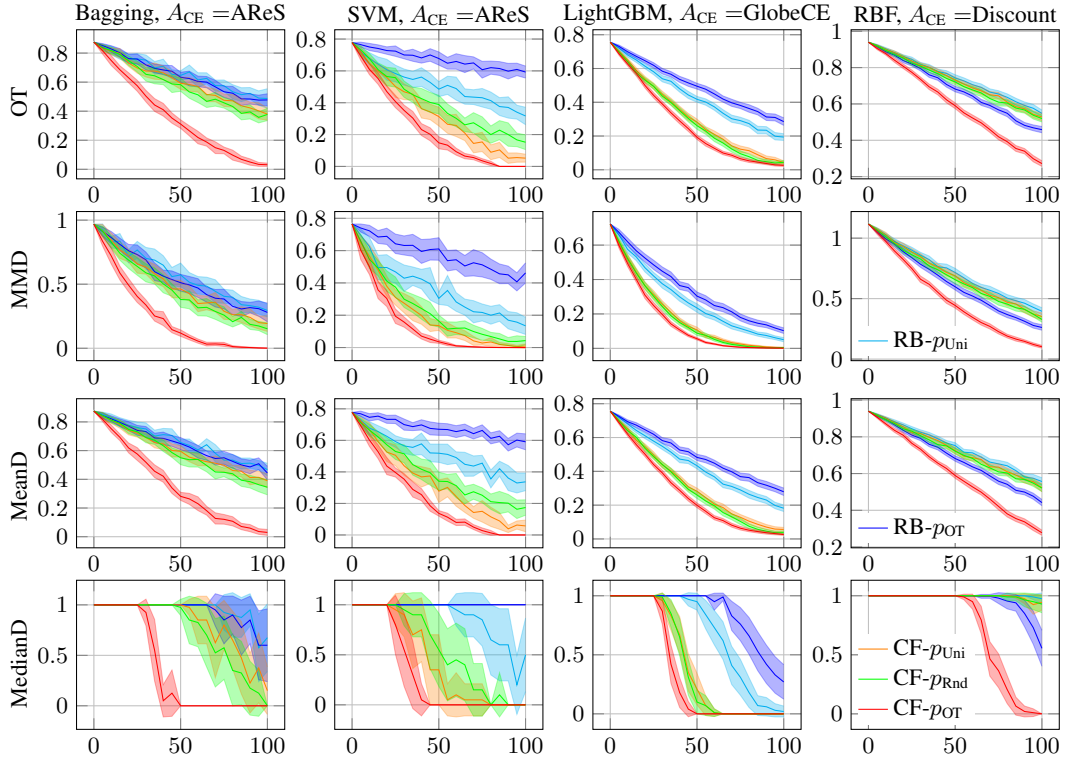


Figure 8: [COMPAS] Extended per-scenario evidence for C2: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. the action budget C , with $\text{CF-}p_{\text{OT}}$ (our p -SHAP) ahead across panels. Experiments use 100 runs; shaded bands are 99.9% confidence intervals; legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.